# Pranav JVS

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2026 | 7.07 |
| Narayana JR College | Class XII (TS State) | 2023 | 97.70% |
| Narayana School | Class X (TS State) | 2021 | 10.00% |

## WORK EXPERIENCE

**Applied AI Intern | Prof. Tammana**

- Developed a Hackathon Evaluation & Verification Engine combining static code analysis, complexity/testing checks, and problem statement coverage mapping to produce evidence backed technical evaluations.
- Built a sandboxed dynamic validation pipeline to safely execute untrusted submissions, exercise API endpoints, verify data flow, and detect hardcoded or non-functional implementations using isolated, resource-limited containers.
- Integrated Gemini based evaluation to generate structured technical assessments with supporting evidence, identify critical red flags, and formulate targeted probing questions for jury review.
- Built an AI Research Navigator over a curated 50-document corpus spanning LLMs, RAG, agents, alignment, and evaluation, using normalized metadata and topic tags to organize technical research resources.

## PROJECTS

**Robust Motion Gated Adaptation (RMGA) for Video Action Recognition**

- Proposed RMGA, a novel test time adaptation framework to improve robustness of video action recognition models under domain shift without requiring retraining.
- Developed spatiotemporal motion masking to suppress static background noise and focus adaptation on dynamic regions.
- Developed entropy-based temporal selection to adapt on high-confidence video segments.
- Evaluated on R3D, MC3, and R(2+1)D using UCF50, improving accuracy over baselines with competitive results.

**Online Auction Website**

- Designed and implemented a normalized relational database schema in MySQL for an online auction system, modeling entities such as users, listings, bids, and transactions with appropriate primary/foreign key relationships.
- Enforced data consistency through constraints and careful schema design to support concurrent auction workflows.
- Developed SQL triggers to enforce business rules, including bid amount validation, prevention of invalid transactions, and maintaining auction state consistency.
- Optimized query structure for efficient bid tracking, ensuring reliable performance for real time auction operations.

**Quality of Life Analysis**

- Cleaned and analyzed cross country quality of life data using Python to identify trends, outliers, and regional differences.
- Performed exploratory data analysis using descriptive statistics, correlation analysis, and visualizations to examine distributions and relationships between quality of life indicators.
- Applied statistical inference techniques, including confidence intervals, hypothesis testing, and the Central Limit Theorem, to evaluate group differences and quantify uncertainty in population estimates.

**Multi-Agent Homeostasis Simulation(Ongoing)**

- Designed a multi-agent system modeling interacting human organ systems as autonomous agents, with each agent maintaining local states, observing physiological variables, and responding to changes in the simulated environment.
- Developed agent decision-making and communication mechanisms to coordinate responses across physiological variables under controlled disturbances, enabling agents to adapt behavior based on shared system information.
- Modeled physiological disturbances and inter-agent dependencies to study how local responses propagate through the system and affect overall homeostasis.
- Evaluated independent, communicating, and cooperating agent configurations under simulated disturbances using system stability, deviation from homeostasis, and recovery behavior as quantitative measures.

## RELEVANT COURSEWORK

**Maths:** Probability and Random Variables, Applied Statistics, Matrix Theory

**AI:** Foundations of Machine Learning, Deep Learning, Computer Vision, Multi Agent Systems

**Software:** DSA, Database and Management Systems

## SKILLS

**Languages:** Python, C++, SQL

**ML/AI:** PyTorch, TensorFlow, Scikit-learn Hugging Face, OpenCV

**GenAI:** LLMs, RAG, AI Agents, LangChain, Prompt Engineering

**Data:** NumPy, Pandas, Matplotlib, Seaborn

**Tools:** Git, Docker, Jupyter, Kaggle

## POSITIONS OF RESPONSIBILITY

**Sunshine Buddy and Mentor**

- Organized departmental meets for batchmates and conducted Mentor-Mentee meetings for juniors to support their transition into the institute.
- Volunteered in Sunshine events, contributing to event coordination and student engagement

**GLITCH Core Member**

- Organized and coordinated gaming events, including game testing and inter-hostel competitions, as part of the core organizing team.